

## Bachelor of Technology — Civil Engineering

Table 1: Civil Engineering — B24 Credit Distribution

| Division           | Subdivision                 | Credits    |
|--------------------|-----------------------------|------------|
| IC Core Courses    | IC Compulsory               | 32         |
|                    | IC Basket                   | 6          |
|                    | IKS + HSS                   | 15         |
| Discipline Courses | Discipline Core             | 49         |
|                    | Discipline Electives        | 21         |
| Electives          | Free Electives              | 25         |
|                    | MTP (4+4, equivalent as DE) | 8          |
|                    | ISTP (equivalent as FE)     | 4          |
| <b>Total</b>       |                             | <b>160</b> |

Table 2: B. Tech Civil Engineering — B24 Discipline Core Courses

| Sl. No       | Course Code | Course Name                                  | L-T-P-C | Credits   |
|--------------|-------------|----------------------------------------------|---------|-----------|
| 1            | CE-202      | Introduction to Civil Engineering Profession | 1-0-0-1 | 1         |
| 2            | CE-203      | Construction Materials                       | 3-0-0-3 | 3         |
| 3            | CE-203P     | Building (Constructions) Materials Lab       | 0-0-2-1 | 1         |
| 4            | CE-252      | Geology and Geomorphology                    | 2-0-2-3 | 3         |
| 5            | CE-310      | Strength of Materials and Structures         | 3-0-0-3 | 3         |
| 6            | CE-310P     | Strength of Materials and Structures Lab     | 0-0-2-1 | 1         |
| 7            | CE-201      | Surveying Traditional and Digital            | 2-0-4-4 | 4         |
| 8            | CE-251      | Hydraulics Engineering                       | 3-0-0-3 | 3         |
| 9            | CE-304P     | Hydraulics Engineering Lab                   | 0-0-2-1 | 1         |
| 10           | CE-311      | Geotechnical Engineering I                   | 3-0-0-3 | 3         |
| 11           | CE-311P     | Geotechnical Engineering Lab                 | 0-0-2-1 | 1         |
| 12           | CE-404      | Analysis of Structures                       | 3-0-0-3 | 3         |
| 13           | CE-303      | Water Resources Engineering                  | 3-0-0-3 | 3         |
| 14           | CE-351      | Design of Reinforced Concrete Structures     | 2-1-0-3 | 3         |
| 15           | CE-352      | Transportation Engineering                   | 3-0-0-3 | 3         |
| 16           | CE-352P     | Transportation Engineering Lab               | 0-0-2-1 | 1         |
| 17           | CE-353P     | Civil Engineering Drawing                    | 0-0-2-1 | 1         |
| 18           | CE-402      | Geotechnical Engineering II                  | 2-1-0-3 | 3         |
| 19           | CE-305P     | Environmental Engineering Lab                | 0-0-2-1 | 1         |
| 20           | CE-356      | Reverse Engineering                          | 1-0-0-1 | 1         |
| 21           | CE-401      | Design of Steel Structures                   | 2-1-0-3 | 3         |
| 22           | CE-403      | Water and Wastewater Engineering             | 3-0-0-3 | 3         |
| <b>Total</b> |             |                                              |         | <b>49</b> |

Table 3: B. Tech Civil Engineering — B24 Semester 1

| Sl. No       | Course Category         | Course Code | Course Name                                          | L-T-P-C     | Credits   |
|--------------|-------------------------|-------------|------------------------------------------------------|-------------|-----------|
| 1            | IC                      | IC-112      | Calculus                                             | 1.5-0.5-0-2 | 2         |
| 2            | IC                      | IC-113      | Complex Variables and Vector Calculus                | 1.5-0.5-0-2 | 2         |
| 3            | IC                      | IC-152      | Introduction to Python and Data Science              | 3-0-2-4     | 4         |
| 4            | IC Basket               | IC-230      | Environmental Science                                | 3-0-0-3     | 3         |
| 5            | IKS + HSS<br>(choose 1) | HS-108      | Basic English for Engineers                          | 3-0-0-3     | 3         |
| 6            |                         | HS-110      | Japanese for Beginners                               | 3           |           |
| 7            |                         | HS-151      | Introduction to English Literature                   | 3-0-0-3     |           |
| 8            |                         | HS-256      | Introduction to Spanish                              | 3           |           |
| 9            | IKS + HSS               | IC-181      | Introduction to Consciousness and Holistic Wellbeing | 2-0-2-3     | 3         |
| <b>Total</b> |                         |             |                                                      |             | <b>17</b> |

*Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.*

Table 4: B. Tech Civil Engineering — B24 Semester 2

| Sl. No       | Course Category | Course Code | Course Name                                  | L-T-P-C     | Credits   |
|--------------|-----------------|-------------|----------------------------------------------|-------------|-----------|
| 1            | IC              | IC-114      | Linear Algebra                               | 2-0-0-2     | 2         |
| 2            | IC              | IC-115      | ODE & Integral Transforms                    | 2-0-0-2     | 2         |
| 3            | IC              | IC-140      | Engineering Graphics for Design              | 2-0-3-4     | 4         |
| 4            | IC              | IC-161      | Applied Electronics                          | 3           | 3         |
| 5            | IC              | IC-161P     | Applied Electronics Lab                      | 2           | 2         |
| 6            | IC              | IC-252      | Probability and Statistics                   | 3-0-2-4     | 4         |
| 7            | IC Basket       | IC-240      | Mechanics of Rigid Bodies                    | 1.5-1.5-0-3 | 3         |
| 8            | DC              | CE-202      | Introduction to Civil Engineering Profession | 1-0-0-1     | 1         |
| <b>Total</b> |                 |             |                                              |             | <b>21</b> |

*Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.*

Table 5: B. Tech Civil Engineering — B24 Semester 3

| Sl. No       | Course Category | Course Code | Course Name                              | L-T-P-C | Credits   |
|--------------|-----------------|-------------|------------------------------------------|---------|-----------|
| 1            | IC              | IC-272      | Machine Learning                         | 2-0-2-3 | 3         |
| 2            | DC              | CE-203      | Construction Materials                   | 3-0-0-3 | 3         |
| 3            | DC              | CE-203P     | Building (Constructions) Materials Lab   | 0-0-2-1 | 1         |
| 4            | DC              | CE-252      | Geology and Geomorphology                | 2-0-2-3 | 3         |
| 5            | DC              | CE-310      | Strength of Materials and Structures     | 3-0-0-3 | 3         |
| 6            | DC              | CE-310P     | Strength of Materials and Structures Lab | 0-0-2-1 | 1         |
| <b>Total</b> |                 |             |                                          |         | <b>14</b> |

*Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.*

Table 6: B. Tech Civil Engineering — B24 Semester 4

| Sl. No       | Course Category | Course Code | Course Name                       | L-T-P-C | Credits   |
|--------------|-----------------|-------------|-----------------------------------|---------|-----------|
| 1            | IC              | IC-222P     | Physics Practicum                 | 0-0-3-2 | 2         |
| 2            | DC              | CE-201      | Surveying Traditional and Digital | 2-0-4-4 | 4         |
| 3            | DC              | CE-251      | Hydraulics Engineering            | 3-0-0-3 | 3         |
| 4            | DC              | CE-304P     | Hydraulics Engineering Lab        | 0-0-2-1 | 1         |
| 5            | DC              | CE-311      | Geotechnical Engineering I        | 3-0-0-3 | 3         |
| 6            | DC              | CE-311P     | Geotechnical Engineering Lab      | 0-0-2-1 | 1         |
| 7            | DC              | CE-404      | Analysis of Structures            | 3-0-0-3 | 3         |
| <b>Total</b> |                 |             |                                   |         | <b>17</b> |

*Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.*

Table 7: B. Tech Civil Engineering — B24 Semester 5

| Sl. No       | Course Category | Course Code | Course Name                              | L-T-P-C | Credits   |
|--------------|-----------------|-------------|------------------------------------------|---------|-----------|
| 1            | DC              | CE-303      | Water Resources Engineering              | 3-0-0-3 | 3         |
| 2            | DC              | CE-351      | Design of Reinforced Concrete Structures | 2-1-0-3 | 3         |
| 3            | DC              | CE-352      | Transportation Engineering               | 3-0-0-3 | 3         |
| 4            | DC              | CE-352P     | Transportation Engineering Lab           | 0-0-2-1 | 1         |
| 5            | DC              | CE-353P     | Civil Engineering Drawing                | 0-0-2-1 | 1         |
| 6            | DC              | CE-402      | Geotechnical Engineering II              | 2-1-0-3 | 3         |
| <b>Total</b> |                 |             |                                          |         | <b>14</b> |

*Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.*

Table 8: B. Tech Civil Engineering — B24 Semester 6

| Sl. No       | Course Category | Course Code | Course Name                                 | L-T-P-C | Credits   |
|--------------|-----------------|-------------|---------------------------------------------|---------|-----------|
| 1            | DC              | CE-305P     | Environmental Engineering Lab               | 0-0-2-1 | 1         |
| 2            | DC              | CE-356      | Reverse Engineering                         | 1-0-0-1 | 1         |
| 3            | DC              | CE-401      | Design of Steel Structures                  | 2-1-0-3 | 3         |
| 4            | DC              | CE-403      | Water and Wastewater Engineering            | 3-0-0-3 | 3         |
| 5            | ISTP            | DP-301P     | Interdisciplinary Socio-Technical Practicum | 0-0-6-4 | 4         |
| <b>Total</b> |                 |             |                                             |         | <b>12</b> |

*Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.*

Table 9: B. Tech Civil Engineering — B24 Semester 7

| Sl. No       | Course Category | Course Code | Course Name                 | L-T-P-C | Credits  |
|--------------|-----------------|-------------|-----------------------------|---------|----------|
| 1            | MTP             | CE-498P     | Major Technical Project - I | 0-0-6-4 | 4        |
| <b>Total</b> |                 |             |                             |         | <b>4</b> |

*Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.*

Table 10: B. Tech Civil Engineering — B24 Semester 8

| Sl. No       | Course Category | Course Code | Course Name                  | L-T-P-C | Credits  |
|--------------|-----------------|-------------|------------------------------|---------|----------|
| 1            | IC              | CE-010      | Internship                   |         | 2        |
| 2            | MTP             | CE-499P     | Major Technical Project - II | 0-0-6-4 | 4        |
| <b>Total</b> |                 |             |                              |         | <b>6</b> |

*Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.*

*Beyond the core courses above, students choose 21 credits of **Discipline Electives** and 25 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 11: Civil Engineering — Available Discipline Electives (choose from these)

| Sl. No | Course Code | Course Name                                     | Credits |
|--------|-------------|-------------------------------------------------|---------|
| 1      | CD-591      | Special Topics in Disaster Resilient Hill Roads | 1       |
| 2      | CE-304      | Hydraulics Engineering Lab                      | 3       |
| 3      | CE-305      | Environment and Earth Science Lab               | 3       |
| 4      | CE-306      | Environmental Engineering Lab                   | 3       |
| 5      | CE-353      | Civil Engineering Drawing                       | 3       |
| 6      | CE-354      | Building and Pavement Materials Laboratory      | 3       |

*Civil Engineering — Available Discipline Electives (choose from these) (continued)*

| Sl. No | Course Code | Course Name                                                    | Credits |
|--------|-------------|----------------------------------------------------------------|---------|
| 7      | CE-355      | Hydrology                                                      | 3       |
| 8      | CE-405      | Water and Wastewater Engineering                               | 3       |
| 9      | CE-451      | Irrigation Engineering and Hydraulic Structures                | 3       |
| 10     | CE-452      | Open Channel Flow                                              | 3       |
| 11     | CE-501      | Remote Sensing                                                 | 3       |
| 12     | CE-502      | Groundwater Flow and Contaminant Transport                     | 3       |
| 13     | CE-503      | Fundamentals of Project Management                             | 3       |
| 14     | CE-504      | Slope Stability and Retaining Structures                       | 3       |
| 15     | CE-505      | Engineering of Ground Modification                             | 3       |
| 16     | CE-506      | Analysis of Indeterminate Structures                           | 3       |
| 17     | CE-507      | Advanced Concrete Science                                      | 3       |
| 18     | CE-508      | Photogeology and Photogrammetry                                | 3       |
| 19     | CE-509      | Bridge Engineering                                             | 3       |
| 20     | CE-510      | Modelling and Simulation in Water Resources Engineering        | 3       |
| 21     | CE-511      | Structural Dynamics with Application to Earthquake Engineering | 3       |
| 22     | CE-512      | Advanced Soil Mechanics                                        | 3       |
| 23     | CE-513      | Advanced Foundation Engineering                                | 3       |
| 24     | CE-514      | Rock Mechanics                                                 | 3       |
| 25     | CE-515      | Environmental Impact Assessment                                | 3       |
| 26     | CE-516      | Uncertainty Analysis in Civil Engineering                      | 3       |
| 27     | CE-517      | Hydroinformatics                                               | 4       |
| 28     | CE-518      | Structural Reliability and Risk Assessment                     | 3       |
| 29     | CE-519      | Chemistry of Natural Waters                                    | 3       |
| 30     | CE-520      | Environmental Reaction Modeling                                | 3       |
| 31     | CE-521      | Ecology and Environment Microbiology                           | 4       |
| 32     | CE-522      | Matrix Methods for Structural Analysis                         | 3       |
| 33     | CE-523      | Building Science                                               | 3       |
| 34     | CE-524      | Applied Hydroclimatology                                       | 3       |
| 35     | CE-524P     | Computational Hydroclimatology Lab                             | 1       |
| 36     | CE-526      | Critical Zone Science                                          | 3       |
| 37     | CE-551      | Geosynthetics and their applications                           | 3       |
| 38     | CE-552      | Concrete Technology                                            | 3       |
| 39     | CE-554      | Prestressed Concrete Structures                                | 3       |
| 40     | CE-555      | Advanced Design of Structures                                  | 3       |
| 41     | CE-556      | Structural Engineering Laboratory                              | 3       |
| 42     | CE-556P     | Structural Engineering laboratory                              | 2       |
| 43     | CE-557      | Solid Mechanics in Structural Engineering                      | 3       |
| 44     | CE-558      | Air pollution and its mitigation                               | 3       |
| 45     | CE-559      | Biological Wastewater Treatment                                | 3       |

*Civil Engineering — Available Discipline Electives (choose from these) (continued)*

| Sl. No                                                               | Course Code | Course Name                                              | Credits |
|----------------------------------------------------------------------|-------------|----------------------------------------------------------|---------|
| 46                                                                   | CE-560      | Soil Dynamics                                            | 3       |
| 47                                                                   | CE-561      | The science of climate change                            | 3       |
| 48                                                                   | CE-562      | Numerical Methods for Engineering Computation            | 3       |
| 49                                                                   | CE-563      | Finite Element Methods in Engineering                    | 4       |
| 50                                                                   | CE-564      | Fundamentals of Fracture Mechanics                       | 3       |
| 51                                                                   | CE-567      | Highway Safety and Modeling                              | 3       |
| 52                                                                   | CE-568P     | Experimental Structural Dynamics                         | 1       |
| 53                                                                   | CE-586      | Mini Project                                             | 3       |
| 54                                                                   | CE-586P     | Mini Project                                             | 3       |
| 55                                                                   | CE-587      | Industrial / Academic Internship                         | 3       |
| 56                                                                   | CE-591      | Special Topics in Civil and Mechanical Engineering       | 3       |
| 57                                                                   | CE-600      | Research Methodology for Civil Engineering               | 1       |
| 58                                                                   | CE-601      | Geo-Informatics                                          | 3       |
| 59                                                                   | CE-602      | Blast Engineering                                        | 3       |
| 60                                                                   | CE-604      | Theory of Plasticity                                     | 3       |
| 61                                                                   | CE-605      | Engineering Seismology and Seismic Hazard Assessment     | 3       |
| 62                                                                   | CE-606      | Constitutive Modeling of Frictional Material             | 3       |
| 63                                                                   | CE-610      | Analysis and Design for Earthquake Resistant Structures  | 3       |
| 64                                                                   | CE-611      | Structural Health Monitoring                             | 3       |
| 65                                                                   | CE-612      | Theory of Plates and Shells                              | 3       |
| 66                                                                   | CE-613      | Mechanics of Unsaturated Soils                           | 3       |
| 67                                                                   | CE-617      | Mechanics of Composite Materials                         | 3       |
| 68                                                                   | CE-688      | Post Graduate Project - I                                | 3       |
| 69                                                                   | CE-689      | Post Graduate Project - II                               | 3       |
| <i>Compulsory project components — counted as DE, not selectable</i> |             |                                                          |         |
| 70                                                                   | CE-498P     | Major Technical Project - I (compulsory; counted as DE)  | 4       |
| 71                                                                   | CE-499P     | Major Technical Project - II (compulsory; counted as DE) | 4       |